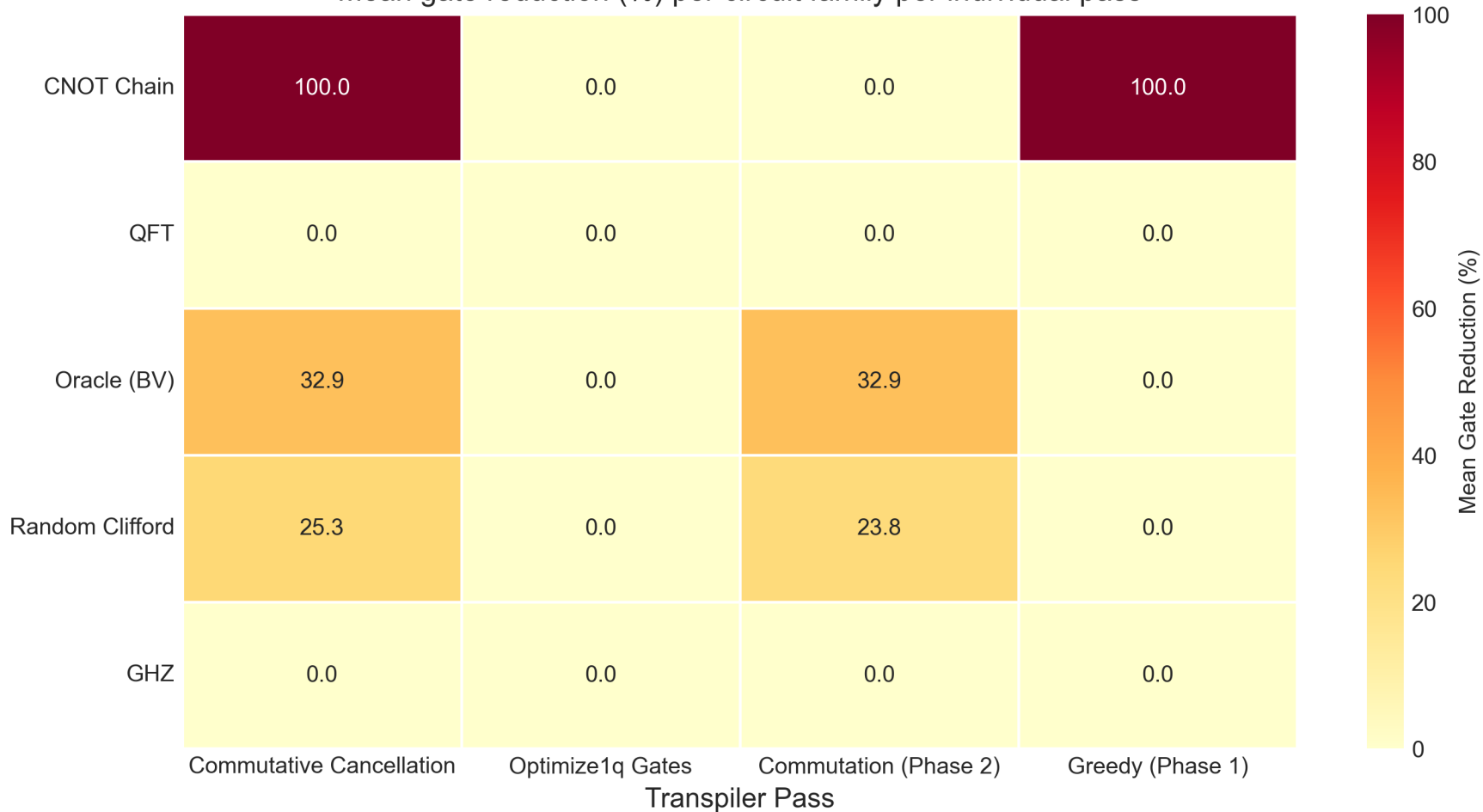


Figure 16: Pass Effectiveness Heatmap
Mean gate reduction (%) per circuit family per individual pass



Key finding: CommutativeCancellation matches our Phase 2 on Oracle/RandomClifford but both miss QFT/GHZ. Neither isolated pass explains the full-pipeline advantage (template matching + routing + phase polynomial).